

A Beginner's Guide to the Linux File System

This is an article targeted at Linux beginners. It helps readers get an understanding of the Linux file system through a few experiments that they, too, could try out.

In Linux, everything is treated as a file. Besides data or program files, the hardware devices are also treated as files. Hard disks (and their partitions), CD-ROM drives, keyboards, mice, terminals, etc., are all files. In the Linux file system, every hardware device is represented by a unique file name. These files are often called *device files* and are stored under the directory `/dev`.

The following experiments demonstrate the concept of device files in Linux. I have tried these on my Ubuntu 14.04 desktop.

Experimenting with the device file representing the mouse

The device file that represents the mouse is `/dev/psaux` (i.e., the file `psaux` in the `/dev` directory).

Step 1: Open a terminal window.

Step 2: Type the following command at the prompt:

```
sudo cat /dev/psaux
```

Step 3: Give the administrator password and the system waits for the input (in this case the `cat` command expects input from the file `/dev/psaux`, which is actually the mouse itself).

Step 4: Now move the mouse, and watch the garbage characters on the screen.

Experimenting with the device file representing the terminal

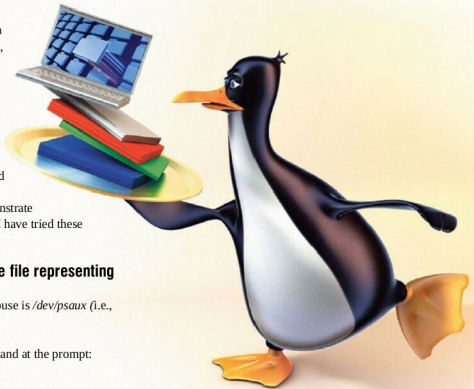
The terminal window, which we open using the *gnome-terminal* program, is actually a software emulation of a hardware device called *Terminal*. That is, this terminal also has an associated device file name in the `/dev` directory.

Step 1: Open two terminal windows.

Step 2: Find the file name associated with each terminal by typing the following command in each terminal:

```
tty
```

Let the file names be `/dev/pts/2` and `/dev/pts/3` for the



first and second terminals, respectively.

Step 3: From the first terminal (in this example, the file name of which is `/dev/pts/2`), type the following command:

```
sudo cat > /dev/pts/3
```

Give the password and type anything (`/dev/pts/3` is the file name associated with the second terminal).

Step 4: Watch the characters that were typed in the first terminal displayed in the second terminal.

Step 5: Try Step 3 from the second terminal *cating* to the file name of the first terminal.

Identifying the type of a file

As the Linux file system has many files that are of different types, it's important to identify the type of a particular file. The file type describes what kind of information it contains or its file format. For example, the file created using *gedit* is an ASCII Text file. The file created using LibreOffice Writer